

Part I.

$$1. \quad E_0 \left[\sum_{t=0}^{\infty} \beta^t (\log C_t - N_t) \right]$$

$$K_{t+1} = (1-\delta) K_t + z_t K_t^\alpha N_t^{1-\alpha} - z_t C_t$$

$$J = E_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t - N_t + \lambda_t (z_t K_t^\alpha N_t^{1-\alpha} - z_t C_t + (1-\delta) K_t - K_{t+1}) \right]$$

F.O.Cs:

$$\frac{\partial J}{\partial C_t} = \beta^t \left(\frac{1}{C_t} - \lambda_t z_t \right) = 0 \quad (1)$$

$$\frac{\partial J}{\partial N_t} = -1 + \lambda_t z_t K_t^\alpha N_t^{-\alpha} (1-\alpha) = 0 \quad (2)$$

$$\frac{\partial J}{\partial K_{t+1}} = -\lambda_t + \beta E_t \lambda_{t+1} \left[\alpha z_{t+1} K_{t+1}^{\alpha-1} N_{t+1}^{1-\alpha} + (1-\delta) \right] = 0 \quad (3)$$

$$\Rightarrow \left\{ \begin{array}{l} C_t = K_t^\alpha N_t^{1-\alpha} (1-\alpha) = \frac{Y_t}{N_t} (1-\alpha) \quad (4) \end{array} \right.$$

$$\frac{1}{C_t z_t} = \beta E_t \frac{1}{C_{t+1} z_{t+1}} \left[\alpha z_{t+1} K_{t+1}^{\alpha-1} N_{t+1}^{1-\alpha} + 1-\delta \right] \quad (5)$$

$$K_{t+1} = (1-\delta) K_t + z_t K_t^\alpha N_t^{1-\alpha} - z_t C_t \quad (6)$$

2. At steady state $z_t = z = 1$.

We have

$$\begin{cases} c = \frac{Y}{N} (1-\alpha) \end{cases} \quad (7)$$

$$\begin{cases} \frac{1}{c} = \beta \frac{1}{c} (\alpha \frac{Y}{K} + 1-\delta) \end{cases} \quad (8)$$

$$\begin{cases} K = (1-\delta)K + Y - c \end{cases} \quad (9)$$

From (8), we get

$$\frac{Y}{K} = \frac{1}{\alpha} \left(\frac{1}{\beta} - 1 + \delta \right) \Rightarrow \frac{K}{Y} = \frac{\alpha}{\frac{1}{\beta} - 1 + \delta}$$

From (9), we can have

$$\frac{c}{Y} = 1 - \delta \frac{K}{Y} = 1 - \frac{\alpha \delta}{\frac{1}{\beta} - 1 + \delta}$$

$$\text{From (7),} \quad N = (1-\alpha) \frac{Y}{c} = \frac{(1-\alpha) \left(\frac{1}{\beta} - 1 + \delta \right)}{\frac{1}{\beta} - 1 + \delta - \alpha \delta}$$

From $Y = C + I$,

$$\frac{I}{Y} = 1 - \frac{c}{Y} = \frac{\alpha \delta}{\frac{1}{\beta} - 1 + \delta}$$

3. For ④, Let $F(c_t, k_t, N_t) \equiv K_t^\alpha N_t^{1-\alpha} c_t^{-1}$

$$F_1 = - \frac{F_{c_t}}{c_t} \Big|_{ss} = - \frac{1}{c}$$

$$F_2 = \frac{\alpha F_{k_t}}{k_t} \Big|_{ss} = \frac{\alpha}{k}$$

$$F_3 = \frac{-\alpha F_{N_t}}{N_t} \Big|_{ss} = \frac{-\alpha}{N}$$

Thus, $-\hat{c}_t + \alpha \hat{k}_t + (-\alpha) \hat{n}_t = 0$

For ⑤, Let $G(c_t, c_{t+1}, z_t, z_{t+1}, k_{t+1}, N_{t+1}) \equiv \beta E_t \frac{c_t z_t}{c_{t+1} z_{t+1}} [\alpha z_{t+1} k_{t+1}^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta]$

$$G_1 = \frac{G_{c_t}}{c_t} \Big|_{ss} = \frac{1}{c}$$

$$G_2 = - \frac{G_{c_{t+1}}}{c_{t+1}} \Big|_{ss} = - \frac{1}{c}$$

$$G_3 = \frac{G_{z_t}}{z_t} \Big|_{ss} = \frac{1}{z}$$

$$G_4 = - \frac{G_{z_{t+1}}}{z_{t+1}} + \beta \alpha \frac{Y_{t+1}}{k_{t+1}} \Big|_{ss} = - \frac{1}{z} + \frac{1 - \beta(1-\delta)}{z} \quad (z=1)$$

$$G_5 = \beta E_t \alpha z_{t+1} \frac{Y_{t+1}}{k_{t+1}} (\alpha-1) \cdot \frac{1}{k_{t+1}} \Big|_{ss} = [1 - \beta(1-\delta)] (\alpha-1) \frac{1}{k}$$

$$G_6 = \beta E_t \alpha (1-\alpha) \frac{Y_{t+1}}{k_{t+1}} z_{t+1} \frac{1}{N_{t+1}} \Big|_{ss} = (1-\alpha) [1 - \beta(1-\delta)] \frac{1}{N}$$

Thus, $\hat{c}_t - \hat{c}_{t+1} + \hat{z}_t - \beta(1-\delta) E_t \hat{z}_{t+1} + (1-\alpha) [1 - \beta(1-\delta)] E_t (\hat{n}_{t+1} - \hat{k}_{t+1}) = 0$

Two ways for ⑥

i> Separately.

$$Y_t = K_t^\alpha N_t^{1-\alpha}$$

$$1 = Y_t^{-1} K_t^\alpha N_t^{1-\alpha}$$

$$e^0 = e^{-\hat{y}_t + \alpha \hat{k}_t + (1-\alpha) \hat{n}_t} \Rightarrow \hat{y}_t = \alpha \hat{k}_t + (1-\alpha) \hat{n}_t$$

$$Y_t = C_t + \frac{I_t}{z_t} \Rightarrow Y(1 + \hat{y}_t) = C(1 + \hat{c}_t) + \frac{I(1 + \hat{i}_t)}{(1 + \hat{z}_t)}$$

$$\Rightarrow Y(1 + \hat{y}_t) = C(1 + \hat{c}_t) + I(1 + \hat{i}_t - \hat{z}_t)$$

$$\Rightarrow \hat{y}_t = \hat{c}_t \frac{C}{Y} + \frac{I}{Y} (\hat{i}_t - \hat{z}_t)$$

$$K_{t+1} = (1-\delta)K_t + I_t \Rightarrow K(1 + \hat{k}_{t+1}) = (1-\delta)K(1 + \hat{k}_t) + I(1 + \hat{i}_t)$$

$$\Rightarrow \hat{k}_{t+1} = (1-\delta)\hat{k}_t + \frac{I}{K} \hat{i}_t$$

ii> From ⑥, let $H(K_t, K_{t+1}, C_t, z_t, N_t) \equiv z_t K_t^\alpha N_t^{1-\alpha} - z_t C_t + (1-\delta)K_t - K_{t+1}$

$$H_1 = \left| z_t \frac{Y_t}{K_t} \alpha + 1 - \delta \right|_{ss} = \frac{1}{\beta}, \quad H_2 = -1$$

$$H_3 = -z_t|_{ss} = -1 \quad H_4 = Y_t - C_t|_{ss} = Y - C$$

$$H_5 = (1-\alpha) \frac{Y}{N}$$

$$\Rightarrow \frac{1}{\beta} K \hat{k}_t - K \hat{k}_{t+1} - C \hat{c}_t + (Y - C) \hat{z}_t + (1-\alpha) Y \hat{n}_t = 0$$

$$\Rightarrow \frac{1}{\beta} \frac{K}{Y} \hat{k}_t - \frac{K}{Y} \hat{k}_{t+1} - \frac{C}{Y} \hat{c}_t + (1 - \frac{C}{Y}) \hat{z}_t + (1-\alpha) \hat{n}_t = 0$$

4. Notice that

$u(C_t, N_t) = \log C_t - N_t$ is a King-Plosser-Rebelo utility function.

$MU_N = -1$ is a negative constant.

A positive shock on Z_t means the technology improvement for producing capital goods, which makes saving more desirable. So the HHs would like to reduce consumption and increase savings and capital accumulations. Thus, we have the increase of output, investment, and decrease of consumption. Since the marginal return from labor increases, while the marginal disutility is constant, the labor supply will increase.

Part II

$$5. \quad K_{t+1} = (1 - d(u_t)) K_t + Z_t (u_t^\alpha K_t^\alpha N_t^{1-\alpha} - C_t)$$

$$J = E_0 \sum_{t=0}^{\infty} \beta^t \left\{ \log C_t - N_t + \lambda_t \left[Z_t (u_t^\alpha K_t^\alpha N_t^{1-\alpha} - C_t) + (1 - d(u_t)) K_t - K_{t+1} \right] \right\}$$

F.O.C.s

$$\frac{\partial J}{\partial C_t} = \beta^t \left(\frac{1}{C_t} - \lambda_t Z_t \right) = 0 \quad (i)$$

$$\frac{\partial J}{\partial N_t} = \beta^t \left(-1 + \lambda_t Z_t (1-\alpha) (u_t K_t)^\alpha N_t^{-\alpha} \right) = 0 \quad (ii)$$

$$\frac{\partial J}{\partial K_{t+1}} = \beta^t \left[-\lambda_{t+1} + \beta E_t \lambda_{t+1} \left(\alpha Z_{t+1} u_{t+1}^\alpha K_{t+1}^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - d(u_{t+1}) \right) \right] = 0 \quad (iii)$$

$$\frac{\partial J}{\partial u_t} = \beta^t \lambda_t \left(\alpha Z_t u_t^{\alpha-1} K_t^\alpha N_t^{1-\alpha} - d'(u_t) K_t \right) = 0 \quad (iv)$$

$$\Rightarrow \begin{cases} C_t = (1-\alpha) u_t^\alpha K_t^\alpha N_t^{1-\alpha} & (v) \\ \alpha Z_t u_t^{\alpha-1} K_t^\alpha N_t^{1-\alpha} - d'(u_t) K_t = 0 & (vi) \\ K_{t+1} = (1 - d(u_t)) K_t + Z_t (u_t^\alpha K_t^\alpha N_t^{1-\alpha} - C_t) & (vii) \\ \frac{1}{C_t Z_t} = \beta E_t \frac{1}{C_{t+1} Z_{t+1}} \left(\alpha Z_{t+1} u_{t+1}^\alpha K_{t+1}^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - d(u_{t+1}) \right) & (viii) \end{cases}$$

6. At steady state, $z_t = 1$, $d(u_t) = \delta$,

$$\begin{cases} C = (1-\alpha) \frac{Y}{N} \\ \alpha \frac{Y}{K^{\alpha}} - d'(u) = 0 \\ K = K - \delta K + Y - C \\ \frac{1}{C} = \beta \left(\frac{1}{C} \mid \alpha \frac{Y}{K} + 1 - \delta \right) \end{cases}$$

$$\Rightarrow \begin{cases} \frac{Y}{K} = \frac{1}{\alpha} \left(\frac{1}{\beta} + \delta - 1 \right) \\ N = (1-\alpha) \frac{Y}{C} \\ u d'(u) = \alpha \frac{Y}{K} = \frac{1}{\beta} + \delta - 1 \\ \frac{C}{Y} = 1 - \delta \frac{K}{Y} \end{cases}$$

$$\text{or } \frac{K}{Y} = \frac{\alpha}{\frac{1}{\beta} + \delta - 1}, \quad \frac{C}{Y} = 1 - \frac{\delta \alpha}{\frac{1}{\beta} + \delta - 1}, \quad \frac{1}{Y} = 1 - \frac{C}{Y} = \frac{\delta \alpha}{\frac{1}{\beta} + \delta - 1}$$

$$u d'(u) = \frac{1}{\beta} + \delta - 1, \quad N = \frac{(1-\alpha) \left(\frac{1}{\beta} + \delta - 1 \right)}{\frac{1}{\beta} + \delta - 1 - \delta \alpha}$$

7.

For viii), let

$$F(c_t, c_{t+1}, z_t, z_{t+1}, u_{t+1}, K_{t+1}, N_{t+1}) = \beta E_t \frac{c_t z_t}{c_{t+1} z_{t+1}} \left(\alpha z_{t+1} u_{t+1}^\alpha K_{t+1}^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - d(u_{t+1}) \right)$$

$$F_1 = \frac{1}{c}, \quad F_2 = -\frac{1}{c}, \quad F_3 = \frac{1}{z}$$

$$F_4 = -\frac{1}{z} + \beta \alpha \frac{Y}{K} \frac{1}{z}, \quad F_5 = \beta \left(\alpha \frac{Y}{K} \frac{1}{u} - \frac{d'(u)u}{u} \right),$$

$$F_6 = \beta \alpha \frac{Y}{K} (1-\alpha) \frac{1}{K}, \quad F_7 = \beta \alpha \frac{Y}{K} (1-\alpha) \frac{1}{N}$$

Thus,

$$0 = \hat{c}_t - \hat{c}_{t+1} + \hat{z}_t + (-1 + \beta \alpha \frac{Y}{K} E_t \hat{z}_{t+1} + \beta \left(\alpha \frac{Y}{K} - d'(u)u \right) E_t \hat{u}_{t+1} + \beta \alpha (1-\alpha) \frac{Y}{K} E_t (\hat{n}_{t+1} - \hat{k}_{t+1})) \quad (ivv)$$

For (vii), let $G(c_t, u_t, K_t, K_{t+1}, N_t, z_t) = z_t \left(u_t^\alpha K_t^\alpha N_t^{1-\alpha} - c_t \right) + (1 - d(u_t)) K_t - K_{t+1}$

$$G_1 = -1, \quad G_2 = \frac{\alpha Y}{u} - \frac{d'(u)u}{u} K, \quad G_3 = \alpha \frac{Y}{K} + (1-\delta), \quad G_4 = -1$$

$$G_5 = (1-\alpha) \frac{Y}{N}, \quad G_6 = Y - C$$

Thus,

$$(Y-C)\hat{z}_t - C\hat{c}_t + (\alpha Y - d'(u)u K)\hat{u}_t + [\alpha Y + (1-\delta)K]\hat{k}_t - K\hat{k}_{t+1} + (1-\alpha)Y\hat{n}_t = 0$$

$$(1 - \frac{C}{Y})\hat{z}_t - \frac{C}{Y}\hat{c}_t + (\alpha - d'(u)u \frac{K}{Y})\hat{u}_t + [\alpha + (1-\delta)\frac{K}{Y}]\hat{k}_t - \frac{K}{Y}\hat{k}_{t+1} + (1-\alpha)\hat{n}_t = 0$$

$$\text{or } (1 - \frac{C}{Y})\hat{z}_t - \frac{C}{Y}\hat{c}_t + [\alpha + (1-\delta)\frac{K}{Y}]\hat{k}_t - \frac{K}{Y}\hat{k}_{t+1} + (1-\alpha)\hat{n}_t = 0 \quad (VV)$$

For vi.

$$\text{Let } H(u_t, k_t, N_t, z_t) \equiv \alpha z_t u_t^{\alpha-1} k_t^{\alpha-1} N_t^{1-\alpha} - d'(u_t)$$

$$H_1 = \alpha \frac{Y}{uK} \frac{1}{u} (\alpha-1) - d''(u)$$

$$H_2 = \alpha \frac{Y}{uK} (\alpha-1) \frac{1}{K}, \quad H_3 = \alpha \frac{Y}{uK} (1-\alpha) \frac{1}{N}, \quad H_4 = \alpha \frac{Y}{Ku} \frac{1}{z}$$

Notice that $\frac{\alpha Y}{Ku} = d'(u)$, we get

$$\left(d'(u) (\alpha-1) - u d''(u) \right) \hat{u}_t + (1-\alpha) d'(u) (\hat{n}_t - \hat{k}_t) + d'(u) \hat{z}_t = 0$$

$$\text{or } \left(\alpha-1 - \frac{d''(u)u}{d'(u)} \right) \hat{u}_t + (1-\alpha) (\hat{n}_t - \hat{k}_t) + \hat{z}_t = 0 \quad (VVi)$$

$$\text{For V. Let } I(c_t, u_t, k_t, N_t) = (1-\alpha) u_t^\alpha k_t^\alpha N_t^{-\alpha} c_t^{-1}$$

$$I_1 = -\frac{1}{c+kN} = -\frac{1}{c}, \quad I_2 = \alpha \frac{1}{u}, \quad I_3 = \alpha \frac{1}{K}, \quad I_4 = -\alpha \frac{1}{N}$$

$$\Rightarrow -\hat{c}_t + \alpha \hat{u}_t + \alpha \hat{k}_t - \alpha \hat{n}_t = 0 \quad (VVii)$$

8. From our log-linearized system, we know only

$\frac{d''(u)u}{d'(u)}$ appears in equation (VVI), while all the other coefficients

have been given at steady state.

Thus, the elasticity of $d'(u)$ evaluated in the steady state is the only new parameter that has to be calibrated to study the log-linearized system.

9. — From equation (VI), we can see that, as z_t increases, the marginal return of capital increases. It's desirable to increase the rate of capital utilization.

— This is because u_t increases a lot, which increases the output high enough to offset the effect of more savings.

— Two channels for the increase of output, investment, and hours worked:
i) the increase of z_t itself increases the technology for producing capital goods, which will increase output, investment and hours worked;
ii) u_t increases because of the positive shock of z_t , which also increases the level of output, investment, and hours worked.